

## GENICO TECHNICAL OPERATIONS MANUAL

# GeniCo AuraPurify Smart HEPA Air Purifier: Manufacturing Assembly & Quality Control Runbook

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A comprehensive guide for manufacturing personnel outlining the step-by-step assembly, testing, and quality control procedures for the GeniCo AuraPurify smart HEPA air purifier.

## 1. Introduction and Safety Protocols

The GeniCo AuraPurify Smart HEPA Air Purifier represents a significant advancement in GeniCo's commitment to enhancing consumer well-being through innovative smart home solutions. This sophisticated device is engineered to deliver superior indoor air quality, integrating a multi-stage filtration system with advanced smart technology for unparalleled performance and user convenience. The AuraPurify features a True HEPA filter, capable of capturing 99.97% of airborne particles as small as 0.3 microns, alongside an activated carbon filter for odor and VOC removal, and a durable pre-filter for larger particulates. Its smart capabilities include seamless integration with the GeniConnect mobile application, voice assistant compatibility (e.g., AuraVoice, GeniHome), real-time air quality monitoring via integrated particulate matter (PM2.5) sensors, and adaptive fan speed control. With a Clean Air Delivery Rate (CADR) of 410 m³/h (Pollen), 380 m³/h (Dust), and 360 m³/h (Smoke), the AuraPurify is designed to effectively purify spaces up to 55 square meters (approximately 592 sq ft), operating with an energy efficiency rating of 0.45 kWh/day and a whisper-quiet minimum noise level of 23 dB(A). The meticulous assembly and rigorous quality control outlined in this runbook are paramount to upholding GeniCo's reputation for excellence and ensuring the AuraPurify delivers its promised performance and reliability.

Adherence to stringent safety protocols is not merely a regulatory requirement but a fundamental pillar of GeniCo's operational philosophy. All personnel involved in the manufacturing and assembly of the AuraPurify Smart HEPA Air Purifier are mandated to comply with the established safety guidelines without exception. A critical aspect of this is **Electrostatic Discharge (ESD) prevention**. Electronic components, particularly the AuraPurify's integrated smart module and sensor arrays, are highly susceptible to damage from ESD events, which can compromise functionality and long-term reliability. Therefore, all workstations within the assembly area are equipped with certified ESD-safe mats, and operators must wear grounded wrist straps at all times when handling sensitive components. Regular calibration and testing of ESD equipment, including wrist straps and grounding points, are conducted weekly by the Quality Assurance team, with records maintained in the plant's digital logbook (Log ID: GCO-MFG-ESD-2026-Q2). Furthermore, sensitive electronic parts must only be removed from their protective, anti-static packaging immediately prior to installation and handled exclusively within designated ESD-protected zones.

Beyond ESD precautions, the proper handling of all components and the consistent use of **Personal Protective Equipment (PPE)** are non-negotiable. Sensitive electronic components, such as printed circuit boards (PCBs), integrated circuits (ICs), and sensor modules, must be handled with care, minimizing direct skin contact. Operators are required to use non-magnetic, anti-static tools for assembly and ensure components are never dropped or subjected to excessive force. All personnel on the assembly line must wear the following mandatory PPE:

- **Safety Glasses:** ANSI Z87.1-compliant, providing impact protection.

- **ESD-Safe Gloves:** To prevent contamination and further mitigate ESD risk.
- **Anti-Static Footwear:** Ensuring continuous grounding and slip resistance.
- **Anti-Static Smocks:** To prevent static charge buildup on clothing.
- **Hearing Protection:** Required in designated high-noise zones (e.g., automated testing booths) where noise levels may exceed 85 dB(A).

Regular training sessions, conducted quarterly by the Plant Safety Officer, reinforce these handling procedures and PPE requirements, ensuring all team members are proficient and compliant.

In the event of an emergency, prompt and coordinated action is essential to ensure the safety of all personnel. All employees must be familiar with the emergency alarm signals and designated evacuation routes. Upon activation of a general emergency alarm (continuous siren), all work must cease immediately, power to equipment should be safely shut down if possible, and personnel must proceed calmly to their nearest designated assembly point. Under no circumstances should individuals attempt to retrieve personal belongings or re-enter the facility until an "all clear" signal is issued by authorized personnel. The following emergency contact information is prominently displayed throughout the facility and should be memorized by all team leads:

| Emergency Service        | Internal Extension | External Contact |
|--------------------------|--------------------|------------------|
| Plant Operations Manager | 4101               | (555) 123-4567   |
| Safety Officer           | 4105               | (555) 123-4568   |
| First Aid Station        | 4199               | (555) 123-4569   |
| Security Control         | 4000               | (555) 123-4570   |
| Emergency Services       | N/A                | 9-1-1            |

Team leaders are responsible for accounting for all their assigned personnel at the assembly points. Maintaining a vigilant and safety-conscious environment is a shared responsibility that safeguards our workforce and ensures the integrity of GeniCo's products.

## 2. Assembly Procedures (Module-by-Module)

This section meticulously details the sequential, module-by-module assembly procedures for the GeniCo AuraPurify Smart HEPA Air Purifier. Adherence to these established protocols is paramount to ensuring optimal product integrity, performance, and longevity. Each step has been engineered for precision and efficiency, leveraging advanced manufacturing techniques and quality assurance checkpoints. All personnel are required to consult the relevant technical drawings and diagrams referenced herein prior to commencing any assembly task.

### 2.1 Chassis Assembly

The assembly process commences with the foundational chassis structure.

1. **Base Plate Preparation:** Position the main chassis base plate (Part No. AP-CHAS-001) on the designated assembly jig. Verify that the alignment pins engage correctly. Refer to *Diagram A.1: Chassis Base Plate Alignment*.
2. **Side Panel Integration:** Secure the four (4) primary side panels (Part No. AP-CHAS-002A/B/C/D) to the base plate using M3 Philips head screws (Part No. FSTN-M3-PH-08). Apply a torque of **1.2 Nm ± 0.1 Nm** to each screw. Ensure all panel interlocking tabs are fully seated.

3. **Rear Access Panel Mounting:** Install the rear access panel frame (Part No. AP-CHAS-003) using M4 hex-head screws (Part No. FSTN-M4-HX-10). Torque setting: **1.8 Nm  $\pm$  0.2 Nm**. This frame provides structural support for the power inlet and primary ventilation grilles.

## 2.2 Fan Unit Integration

The centrifugal fan unit, critical for air circulation, is integrated following the chassis's structural completion.

1. **Fan Housing Placement:** Carefully position the pre-assembled centrifugal fan housing (Part No. AP-FAN-001) into the designated cavity within the chassis. Ensure the vibration dampening mounts (Part No. AP-FAN-002) are correctly seated in their respective receptacles. Refer to *Diagram A.2: Fan Unit Mounting Points*.
2. **Securing the Fan Unit:** Fasten the fan housing using four (4) M4 anti-vibration screws (Part No. FSTN-M4-AV-12). Apply a torque of **1.5 Nm  $\pm$  0.15 Nm**. Over-tightening can compromise vibration dampening.
3. **Power Cable Routing:** Route the fan motor's power cable (Part No. CBL-FAN-PWR-01) along the designated internal cable management channel, securing it with three (3) Type-C cable clips (Part No. CBL-CLIP-C-03) to prevent interference with other components or moving parts. Connect the 4-pin fan connector to the designated port on the main control board's harness, ensuring a positive lock.

## 2.3 Filter Housing Installation

Proper installation of the multi-stage filter system is essential for the AuraPurify's core function.

1. **HEPA Filter Frame:** Insert the primary HEPA filter frame (Part No. AP-FILT-HEPA-01) into its dedicated slot within the chassis, ensuring the gasket (Part No. AP-FILT-GASK-01) is fully compressed to prevent air bypass. The frame should slide in smoothly until it clicks into place. Refer to *Diagram A.3: Filter Housing Assembly*.
2. **Activated Carbon Pre-Filter Tray:** Position the activated carbon pre-filter tray (Part No. AP-FILT-CARB-01) directly above the HEPA filter frame. Verify that the directional arrows on the pre-filter align with the indicated airflow direction within the chassis. This tray is secured by two (2) quick-release latches (Part No. FSTN-QR-LATCH-02).

## 2.4 Sensor Array Placement

Accurate environmental monitoring is achieved through precise sensor array installation.

1. **PM2.5 Sensor Module:** Mount the PM2.5 particulate matter sensor module (Part No. AP-SENS-PM25-01) onto its dedicated bracket on the internal front panel, using two (2) M2 self-tapping screws (Part No. FSTN-M2-ST-05). Torque setting: **0.4 Nm  $\pm$  0.05 Nm**. Connect its 3-pin data cable (Part No. CBL-SENS-PM25-01) to the sensor hub. Refer to *Diagram A.4: Sensor Array Layout*.
2. **VOC and Temp/Humidity Sensor:** Install the combined VOC (Volatile Organic Compound) and Temperature/Humidity sensor (Part No. AP-SENS-VOC-TH-01) into its shielded enclosure near the air intake. Secure with one (1) M2.5 machine screw (Part No. FSTN-M25-MS-06). Torque setting: **0.6 Nm  $\pm$  0.05 Nm**. Route its 5-pin data cable (Part No. CBL-SENS-VOC-01) to the sensor hub, ensuring it does not obstruct airflow.

## 2.5 Control Board and Wi-Fi Module Installation

The final stage involves integrating the core electronic intelligence of the AuraPurify unit.

1. **Main Control Board Mounting:** Carefully align the main Printed Circuit Board (PCB) assembly (Part No. AP-PCB-MAIN-01) with the standoffs on the chassis's internal rear panel. Secure the PCB using four (4) M3 nylon stand-off screws (Part No. FSTN-M3-NY-08). Hand-tighten until snug, avoiding excessive force to prevent PCB flex. Refer to *Diagram A.5: Main PCB & Wi-Fi Module Installation*.

- 2. **Cable Connections:** Connect all previously routed cables (fan power, sensor data, power inlet) to their respective labeled ports on the main PCB. Verify each connection is firm and correctly oriented.
- 3. **Wi-Fi Module Integration:** Attach the Wi-Fi communication module (Part No. AP-MOD-WIFI-01) to its dedicated M.2 slot on the main PCB. Secure it with the single M2 retention screw (Part No. FSTN-M2-RT-03). Connect the internal antenna cable (Part No. CBL-ANT-WIFI-01) to the module's U.FL connector, ensuring a positive click. Route the antenna cable to the designated internal antenna mounting point on the top chassis panel.
- 4. **Initial Power-Up Check:** Prior to final enclosure, a preliminary power-up test is conducted to verify basic functionality and LED indicator operation. This step is detailed in Section 3.1: Initial Power-On Diagnostics.

3. Quality Control and Functional Testing

GeniCo's unwavering commitment to product excellence is underscored by a rigorous, multi-stage Quality Control (QC) and functional testing protocol for the AuraPurify Smart HEPA Air Purifier. This comprehensive strategy ensures that every unit leaving our manufacturing facilities adheres to the highest standards of performance, reliability, and user experience. QC checkpoints are strategically integrated at each major assembly stage, commencing with **Incoming Component Inspection**, where critical components such as the HEPA filter media (checked for structural integrity and particulate retention efficiency via random sampling), fan motor assemblies (inspected for bearing play and blade balance), and printed circuit boards (PCBs) undergo visual defect screening and basic electrical continuity tests. This initial phase prevents the integration of substandard materials into the production line. Subsequently, **Sub-Assembly QC** verifies the functionality of modular units, including the fan module's operational stability and the sensor module's baseline output linearity, prior to their integration into the main chassis. Finally, **Final Assembly QC** encompasses a thorough visual inspection of the enclosure for cosmetic imperfections, tactile verification of control buttons, and confirmation of LED indicator functionality, ensuring aesthetic and ergonomic integrity before proceeding to comprehensive functional testing.

Upon completion of physical assembly, each AuraPurify unit undergoes a series of advanced functional tests designed to validate its operational parameters and smart capabilities. The process initiates with a **Power-On Self-Test (POST)**, where the embedded system microcontroller executes a diagnostic routine to verify RAM integrity, ROM checksum, and the successful initialization of all critical peripherals (e.g., UART, SPI, I2C buses). A successful POST is indicated by a specific diagnostic LED sequence and the absence of internal error codes, confirming core system health. Following POST, **Fan Speed Calibration** is performed in a controlled acoustic environment. An optical tachometer measures the fan's Revolutions Per Minute (RPM) at each predefined operational setting (Low, Medium, High, Turbo). Units are calibrated to ensure RPMs fall within a stringent tolerance, as detailed below, guaranteeing optimal airflow and acoustic performance. Concurrently, **Sensor Accuracy Validation** is conducted within a sealed, environmentally controlled chamber. The PM2.5 laser-scattering sensor is exposed to a certified 50 µg/m³ PM2.5 reference aerosol, while the VOC metal oxide semiconductor sensor is challenged with 10 ppm ethanol vapor. The integrated temperature and humidity sensors are validated against calibrated reference instruments maintaining 22.0°C and 50.0% RH. Each sensor's output must fall within specified tolerance ranges to ensure precise environmental monitoring.

| Test Parameter  | Reference Value | Acceptable Tolerance Range | Pass/Fail Criteria                    |
|-----------------|-----------------|----------------------------|---------------------------------------|
| Fan Speed (RPM) |                 |                            |                                       |
| Low             | 850 RPM         | 825 - 875 RPM              | Measured RPM within ±25 RPM of target |

|                       |                |                 |                                       |
|-----------------------|----------------|-----------------|---------------------------------------|
| Medium                | 1400 RPM       | 1360 - 1440 RPM | Measured RPM within ±40 RPM of target |
| High                  | 2100 RPM       | 2040 - 2160 RPM | Measured RPM within ±60 RPM of target |
| Turbo                 | 2800 RPM       | 2720 - 2880 RPM | Measured RPM within ±80 RPM of target |
| PM2.5 Sensor Accuracy | 50 µg/m³       | 48 - 52 µg/m³   | Reading within ±2 µg/m³ of reference  |
| VOC Sensor Accuracy   | 10 ppm Ethanol | 9 - 11 ppm      | Reading within ±1 ppm of reference    |
| Temperature Sensor    | 22.0°C         | 21.5 - 22.5°C   | Reading within ±0.5°C of reference    |
| Humidity Sensor       | 50.0% RH       | 49.0 - 51.0% RH | Reading within ±1.0% RH of reference  |

The final phase of functional testing focuses on the AuraPurify's advanced connectivity and smart features. A **Wi-Fi Connectivity Test** ensures the unit's 2.4GHz 802.11n module can successfully detect and connect to a factory-simulated network (SSID: GeniCo\_TestNet, WPA2-PSK security) within 15 seconds, acquire a DHCP IP address, and successfully exchange data packets with a local server. This validates the network hardware and firmware's robust operation. Simultaneously, **Smart Feature Verification** is conducted through an automated system that simulates interactions with the GeniCo Home application and integrated voice command platforms. This involves programmatic commands for power cycling, mode changes (e.g., Auto, Sleep), fan speed adjustments, and scheduling functions via the local network API. For voice command verification, simulated commands such as "Turn on AuraPurify" or "Set fan to medium" are issued via a GeniCo Voice Gateway interface, with the unit's response and operational state meticulously monitored. A unit passes this stage if all simulated commands are executed correctly and promptly, typically within a 2-second response window, confirming seamless integration with GeniCo's smart ecosystem. Any deviation from these stringent criteria results in immediate quarantine for further diagnosis and rework, upholding GeniCo's commitment to delivering unparalleled product quality.

4. Packaging, Labeling, and Shipping Guidelines

The final stages of manufacturing, encompassing packaging, labeling, and shipping, are critically important for ensuring the GeniCo AuraPurify Smart HEPA Air Purifier (Model AP-2000S) reaches the end-user in pristine condition, fully compliant with all regulatory requirements, and reflective of GeniCo's commitment to quality. Adherence to these guidelines safeguards product integrity, minimizes transit damage, and upholds our brand reputation. Any deviation from these established procedures must be formally documented and approved by the Plant Operations Manager.

4.1 Final Packaging Process

- Each AuraPurify AP-2000S unit must be meticulously prepared for shipment according to the following sequence:
- Primary Unit Placement:** The fully assembled and tested AuraPurify AP-2000S unit, with its pre-installed HEPA filter (GNC-AP2000S-HF-V3.0) securely seated and protected by its sealed plastic bag, shall be carefully lowered into the custom-designed primary corrugated carton (Part No. GNC-AP2000S-PC-V1.2). The unit must be oriented correctly, with the control panel facing upwards, aligning with the carton's internal markings.

2. **Protective Inserts:** Custom-molded Expanded Polystyrene (EPS) end-caps (Part No. GNC-AP2000S-EP-V1.0) are to be placed at the top and bottom of the unit within the carton. These inserts provide essential shock absorption and prevent movement during transit. Ensure a snug fit without applying excessive pressure that could deform the unit's casing.

3. **Accessory Placement:**

- The *User Manual* (Part No. GNC-AP2000S-UM-V2.1) and *Quick Start Guide* (Part No. GNC-AP2000S-QSG-V1.1) are to be placed face-up in the designated compartment within the top EPS insert.
- The 1.8-meter grounded power cord (Part No. GNC-PC-US-1.8M), neatly coiled and secured with a reusable hook-and-loop tie, shall be placed in the secondary compartment adjacent to the documentation.
- The infrared remote control (Part No. GNC-AP2000S-RC), sealed in a protective polybag, is to be placed in its dedicated recess within the top EPS insert.

4. **Carton Sealing:** Once all components are correctly placed, the carton flaps must be closed and sealed using high-tensile, 50mm wide polypropylene adhesive tape. A minimum of three overlapping strips must be applied across the main seam and one strip across each side seam to ensure robust closure and tamper resistance.

## 4.2 Labeling Requirements

Accurate and complete labeling is paramount for traceability, regulatory compliance, and efficient logistics. All labels must be applied squarely, without wrinkles or bubbles, and in the specified locations.

1. **Primary Carton Label:** A pre-printed adhesive label (Part No. GNC-AP2000S-CL-V4.0) must be affixed to the largest side of the primary carton. This label shall include:

- **Product Name:** GeniCo AuraPurify Smart HEPA Air Purifier
- **Model Number:** AP-2000S
- **Serial Number:** A unique 16-character alphanumeric identifier (e.g., AP2000S-260618-00001) with a corresponding 128-bit barcode, generated and applied at the final assembly station.
- **Batch/Lot Number:** (e.g., L260618A)
- **Manufacturing Date:** YYYY-MM-DD format (e.g., 2026-06-18)
- **Regulatory Marks:** Clearly visible CE, FCC, RoHS, UL, and WEEE compliance symbols.
- **Country of Origin:** "Made in Malaysia"
- **Handling Symbols:** Standard ISO 780 graphical symbols for "Fragile," "This Way Up," and "Keep Dry."

2. **Shipping Labels:** For outgoing shipments, carrier-specific shipping labels (e.g., FedEx, UPS, DHL) generated by the logistics department will be applied directly adjacent to the primary carton label. These labels must not obscure any regulatory marks or product information.

## 4.3 Handling Procedures for Finished Goods

Proper handling of packaged AuraPurify units is essential to prevent damage during warehousing and transit.

1. **Palletization:** Packaged units are to be stacked onto standard 48"x40" GMA pallets. A maximum of five (5) cartons high is permitted per pallet to maintain structural integrity and prevent crushing. Each layer must be interlocked for stability. Once stacked, the entire pallet must be securely shrink-wrapped with 80-gauge clear polyethylene film, ensuring all cartons are tightly bound. Corner boards are to be applied to vertical edges for added protection.

2. **Warehousing:** Palletized goods must be stored in a climate-controlled environment, maintaining a temperature range of 15-25°C and relative humidity between 40-60%. Pallets should be stored on designated racking, ensuring

adequate clearance from walls and ceilings. The First-In, First-Out (FIFO) principle must be strictly adhered to for inventory rotation.

**3. Loading and Transit:** When loading onto transport vehicles, forklifts must be operated by certified personnel only. Pallets must be positioned securely, utilizing load bars, straps, or airbags to prevent shifting during transit. Drivers are to be instructed on the fragile nature of the cargo and the importance of smooth acceleration, braking, and cornering. Any evidence of mishandling or damage upon receipt at the distribution center must be immediately reported and documented.

#### 4.4 Final Inspection Checklist Before Shipping

Before any palletized goods are released for shipping, a final quality assurance check must be performed by an authorized QA technician. This checklist ensures all packaging and labeling standards have been met.

| Item                                    | Requirement/Standard                                                                                                            | Inspection Result (Pass/Fail) | Inspector ID | Date/Time |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-------------------------------|--------------|-----------|
| <b>1. Primary Carton Integrity</b>      | No visible dents, tears, punctures, or signs of moisture damage.                                                                |                               |              |           |
| <b>2. Carton Sealing</b>                | Three overlapping tape strips across main seam, one on each side seam; tape firmly adhered, no lifting.                         |                               |              |           |
| <b>3. Protective Inserts</b>            | EPS end-caps correctly placed, unit snug within carton, no excessive movement.                                                  |                               |              |           |
| <b>4. Accessory Placement</b>           | Manual, QSG, Power Cord, Remote Control present in correct compartments/recesses, secured.                                      |                               |              |           |
| <b>5. Primary Carton Label</b>          | Correct model (AP-2000S), legible serial number (e.g., AP 2000S-260618-00001), batch, date, regulatory marks, handling symbols. |                               |              |           |
| <b>6. Shipping Label (if applied)</b>   | Correct destination, legible, not obscuring primary label.                                                                      |                               |              |           |
| <b>7. Palletization (if applicable)</b> | Max 5 cartons high, interlocked, shrink-wrapped securely, corner boards present.                                                |                               |              |           |
| <b>8. Overall Presentation</b>          | Clean, professional appearance, ready for customer receipt.                                                                     |                               |              |           |

